

What Hackers Actually Target in Wi-Fi

- Hackers do not directly see or steal the Wi-Fi password
- They target the authentication communication
- Focus is on data exchanged during connection
- The goal is to capture encrypted proof of authentication



Authentication Process

- Device requests connection to the Wi-Fi router
- Router verifies the password
- Secure communication channel is created
- An authentication exchange occurs during this process

What is a Handshake & Why It Matters

- A handshake is an authentication exchange between device and router
- It proves that the correct password is used
- The password itself is never visible
- Capturing the handshake is essential for cracking attempts



Monitor Mode Concept

- Normal mode allows connection to one network only
- Monitor mode allows listening to all nearby Wi-Fi traffic
 - Hackers use it to observe wireless packets
- No connection to the target network is required

Passive vs Active Attacks

Passive Attack:

- Listening to network traffic silently
- No direct interaction with users

Active Attack:

- Interacting with the network
- Forcing devices to reconnect to capture data

Dictionary Attack

- Uses a predefined list of common passwords
- Tries passwords one by one
- Works well against weak and common passwords
- Faster than brute force attacks

Brute Force Attack

- Tries all possible password combinations
- Time-consuming and resource-intensive
- Strong passwords make this attack impractical
- Often blocked by modern security mechanisms

Why Weak Passwords Are Easily Cracked?

- Short password length
- Use of common words or names
- Predictable patterns (123456, password@123)
- Reusing passwords across networks

Role of Encryption **(WPA / WPA2 / WPA3)**

- Encryption protects Wi-Fi communication
- WPA improved security over WEP
- WPA2 uses strong AES encryption
- WPA3 adds protection against brute-force attacks
- Stronger encryption increases attack difficulty